

TEST CASES DOCUMENT

Project Title

AI-Powered Ticket-to-FAQ Pipeline

Test Case 1: Submit Customer Ticket

Test Case ID: TC_001

Module: Customer Portal

Objective: Verify that a customer can submit a support ticket.

Input:

- Subject: VPN Connection Error
- Description: Unable to connect to VPN.

Expected Result:

- Ticket is successfully created.
- Ticket ID is generated.

Status: Pass

Test Case 2: Close Support Ticket

Test Case ID: TC_002

Module: Support Panel

Objective: Verify that support staff can close a ticket.

Input:

- Ticket ID: 101
- Resolution: Restart VPN service

Expected Result:

- Ticket status changes to Closed.
- Resolution is stored.

Status: Pass

Test Case 3: View Closed Tickets

Test Case ID: TC_003

Module: Closed Tickets

Objective: Verify that closed tickets are displayed correctly.

Input:

- Open Closed Tickets page.

Expected Result:

- Closed ticket list is displayed.

Status: Pass

Test Case 4: Generate Clusters

Test Case ID: TC_004

Module: Clustering Engine

Objective: Verify clustering of similar tickets.

Input:

- Closed tickets dataset

Expected Result:

- Similar tickets grouped into clusters.

Status: Pass

Test Case 5: Generate FAQ Draft

Test Case ID: TC_005

Module: Gemini AI FAQ Generator

Objective: Verify FAQ generation from ticket cluster.

Input:

- Cluster containing similar issues.

Expected Result:

- FAQ draft generated successfully.

Status: Pass

Test Case 6: Approve FAQ Draft

Test Case ID: TC_006

Module: Admin Review

Objective: Verify admin approval process.

Input:

- FAQ Draft

Expected Result:

- Status changes to Approved.

Status: Pass

Test Case 7: Publish FAQ

Test Case ID: TC_007

Module: Published FAQs

Objective: Verify FAQ publishing functionality.

Input:

- Approved FAQ

Expected Result:

- FAQ appears in Published FAQs page.

Status: Pass

Test Case 8: Search Published FAQ

Test Case ID: TC_008

Module: Published FAQs

Objective: Verify FAQ search functionality.

Input:

- Search keyword: VPN

Expected Result:

- Matching FAQ displayed.

Status: Pass

Test Case 9: Export FAQs

Test Case ID: TC_009

Module: Published FAQs

Objective: Verify CSV and Markdown export.

Input:

- Click Export button

Expected Result:

- File downloaded successfully.

Status: Pass

Test Case 10: Dashboard Statistics

Test Case ID: TC_010

Module: Dashboard

Objective: Verify dashboard metrics.

Input:

- Open Dashboard

Expected Result:

- Total Tickets displayed.
- Closed Tickets displayed.
- Clusters displayed.
- FAQ Drafts displayed.
- Published FAQs displayed.

Status: Pass

Test Summary

Test Case ID	Module	Result
TC_001	Customer Portal	Pass

Test Case ID	Module	Result
TC_002	Support Panel	Pass
TC_003	Closed Tickets	Pass
TC_004	Clustering Engine	Pass
TC_005	FAQ Generator	Pass
TC_006	Admin Review	Pass
TC_007	Published FAQs	Pass
TC_008	FAQ Search	Pass
TC_009	Export FAQs	Pass
TC_010	Dashboard	Pass

Conclusion

All test cases were executed successfully. The system performed as expected, including ticket submission, clustering, AI-based FAQ generation, admin review, publishing, searching, and exporting FAQs.